

Experiment No. 5

Aim

Hands-on Solidity programming assignments for creating and working with Smart Contracts.

Theory

1. Primitive Data Types, Variables, and Functions (Pure, View)

- **Primitive Data Types:** Solidity includes basic data types such as `uint/int` (unsigned and signed integers of varying sizes like `uint256` and `int128`), `bool` (boolean values such as `true` or `false`), `address` (20-byte Ethereum account or contract address), and `bytes/string` (for binary and textual data).
- **Variables:** Solidity variables consist of:
 - **State variables:** Permanently stored on the blockchain.
 - **Local variables:** Temporary variables created during function execution.
 - **Global variables:** Predefined variables that provide blockchain-related information such as `msg.sender`, `msg.value`, and `block.timestamp`.
- **Functions (Pure, View):**
 - **Pure functions:** Neither read from nor modify the blockchain state.
 - **View functions:** Read state variables but cannot modify them.

These function types help optimize gas consumption and maintain contract integrity.

2. Inputs and Outputs to Functions

- Functions accept **input arguments** to pass data into the contract and return one or more **output values** after executing the required logic.
- Solidity supports **named return variables**, which improve code readability and make the program easier to understand and debug.
- Function parameters can be used to perform calculations, update contract data, or control the execution of specific operations.

3. Visibility, Modifiers, and Constructors

- **Visibility:** Visibility specifies how and from where functions and variables can be accessed.
 - **public:** Accessible both internally within the contract and externally.
 - **private:** Accessible only within the contract where it is defined.
 - **internal:** Accessible within the contract and derived/child contracts.
 - **external:** Mainly callable from external accounts or other contracts.
- **Modifiers:** Modifiers are reusable blocks of code that alter the behavior of functions. They are commonly used for access-control mechanisms, such as restricting a function so that only the contract owner can execute it.
- **Constructors:** A constructor is a special function that is executed only once when a smart contract is deployed. It is generally used to initialize state variables and set the initial state of the contract.

4. Control Flow: If-Else and Loops

- **If-Else:** Conditional statements allow the contract to make decisions based on specific conditions. For example, an `if-else` statement can be used to check whether a user's balance is sufficient before performing a transaction.
- **Loops:** Solidity supports looping constructs such as `for`, `while`, and `do-while`, which allow a block of code to be executed repeatedly. Loops must be used carefully in smart contracts because excessive iterations can increase gas consumption and may cause a transaction to fail due to insufficient gas.

5. Data Structures: Arrays, Mappings, Structs, and Enums

- **Arrays:** Arrays are ordered collections of elements. They can be either fixed-size or dynamic-size.
- **Mappings:** Mappings are key-value data structures that provide efficient lookup of values using a unique key. For example, `mapping(address => uint)` can be used to store balances associated with Ethereum addresses. Mappings do not support direct iteration over their keys.
- **Structs:** Structs are custom user-defined data types that group related variables into a single structure. They are useful for representing complex entities such as students, employees, or products.

- **Enums:** Enums are user-defined types consisting of a fixed set of predefined constants. They improve code readability and are useful for representing different states of an object or process.

6. Data Locations

Solidity provides different data locations for storing and passing data:

- **Storage:** Permanent data location where data is stored on the blockchain. State variables are stored in storage.
- **Memory:** Temporary data location that exists only during function execution. Data stored in memory is cleared after the function finishes execution.
- **Calldata:** A non-modifiable and non-persistent data location used primarily for parameters of external functions. It is generally more gas-efficient than **memory** when passing external function arguments.

7. Transactions: Ether & Wei, Gas & Gas Price, and Sending Transactions

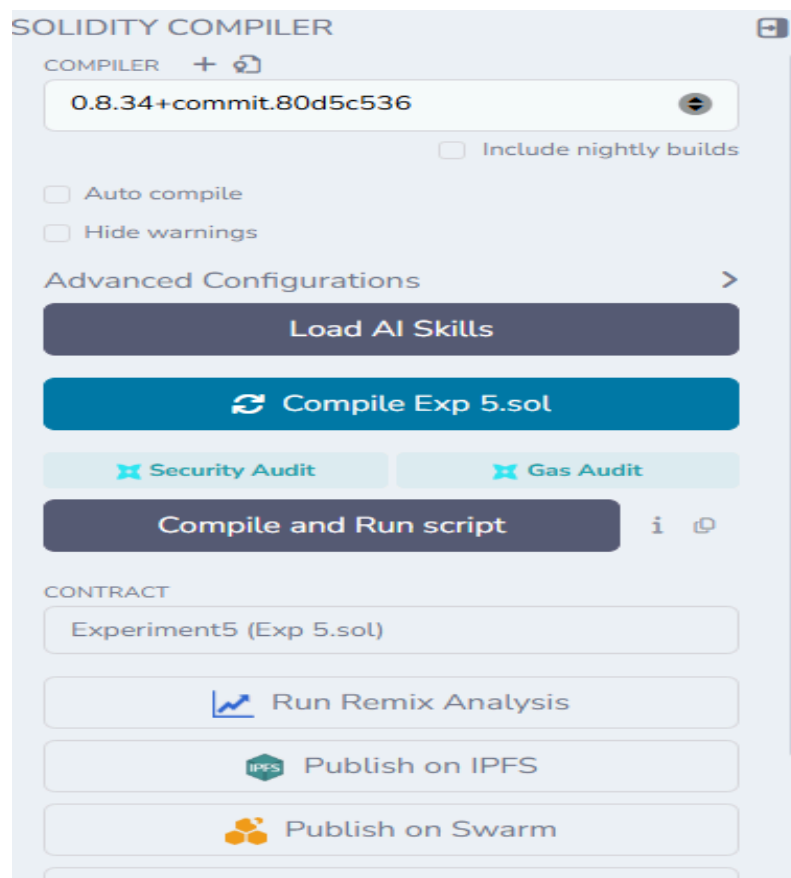
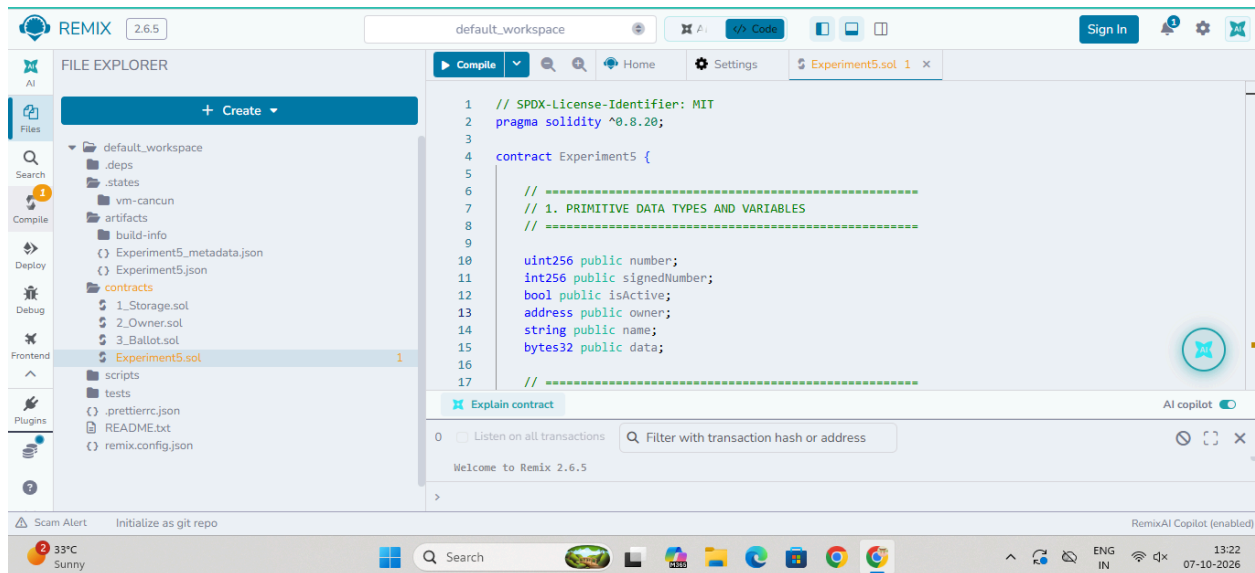
- **Ether and Wei:** Ether is the native cryptocurrency of the Ethereum blockchain. Ether values are internally represented in **Wei**, which is the smallest unit of Ether.

`1 Ether = 1018 Wei`

Using Wei allows Ethereum to represent very small monetary values accurately.

- **Gas and Gas Price:** Gas represents the amount of computational work required to execute an operation or transaction on the Ethereum network. The **gas price** represents the amount of Ether paid for each unit of gas. The total transaction fee depends on the gas consumed and the gas price.
- **Sending Transactions:** Transactions are used to transfer Ether or execute state-changing functions of a smart contract. Solidity provides mechanisms such as `transfer()`, `send()`, and `call()` for transferring Ether between addresses.

OUTPUT



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default_workspace

AI

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DEPLOY & RUN TRANSACTIONS

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Account 1

0x5B3...eddC4

89.999 ETH

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Deployed Contracts

+ Add Contract

Interact with a deployed contract

Experiment5

0xD7A...F771B

Remix VM

28min ago

Functions

Low level interaction

calldata

Value

10

ether

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DEPLOY & RUN TRANSACTIONS

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0x5B3...eddC4

89.999 ETH

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Deployed Contracts 1

Transaction History 9

Deployed Contracts 1

+ Add Contract

Interact with a deployed contract

Experiment5

0xD7A...F771B

Remix VM

30min ago

Functions

+

Low level interaction

-

calldata

Value

10

ether

Compile

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Exp 5.sol

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 contract Experiment5 {
5
6     // =====
7     // 1. Primitive Data Types
8     // =====
9
10    uint public number = 10;
11    int public signedNumber = -5;
12    bool public isActive = true;
13
14    address public owner;
15    string public studentName = "Siddharth";
16    bytes32 public data = "Solidity";
17
18    // =====
19
20 }
```

Explain contract

AI copilot

0

☐ Listen on all transactions

command line interface

- Right-click on a JavaScript file in the file explorer and then click "Run"

The following libraries are accessible:

- ethers.js

Type the library name to see available commands.

>

Alert Initialize as git repo

REMIX

default_workspace

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DEPLOY & RUN TRANSACTIONS

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Cancun

Account 1

0x5B3...eddC4

89.999 ETH

Deploy

Deployed Contracts 1

Transaction History 13

Deployed Contracts 1

+ Add Contract

Interact with a deployed contract

Experiment5

0xD7A...F771B

Remix VM

32min ago

Functions

+

Search functions...

addToArray (uint256)

30

Calldata

Params

Auto-Fill with AI Samples

Transact

approveStudent

Transact

depositBalance (uint256)

100

Calldata

Params

Auto-Fill with AI Samples

Transact

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Exp 5.sol

```
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16    bytes32 public data = "Solidity";
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18    // =====
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20 }
```

Explain contract

AI copilot

0

☐ Listen on all transactions

data: 0xb50...8ba31

Debug

[VM] from: 0x5B3...eddC4 to: Experiment5.(constructor) value: 0 wei data: 0x608...20033 logs: 0

hash: 0xeb6...598ed

Debug

status

1 Transaction mined and execution completed

transaction hash

0xeb6dc95e9e2d03320cd3febfae648db9f19e4c13fd74d241de7de1858a598ed

block hash

0xbe8efbe4c34e1f5287353d5a3014082590fa8b988e64898623c35a5a5aed7bb6

block number

2

contract address

0xd8b934580fce35a11858c6d73ade468a2833fa8

from

0x5B380a6a701c568545dcfc803fc8875f56beddc4

to

Experiment5.(constructor)

transaction cost

1208674 gas

Alert Initialize as git repo

Low level interaction

callldata

Value

10

ether

Gas limit

auto

Transact

wei

gwei

finney

ether

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Account 1

0x5B3...eddC4

89.999 ETH

Experiment5 tx: 0x5db...fbb0a	deploy 35min ago
Experiment5 tx: 0xe0d...4a3cd	deploy 35min ago
Experiment5 tx: 0xeb6...598ed	deploy 37min ago
Experiment5 tx: 0x006...fe525	deploy 37min ago
Experiment5 tx: 0x4d6...dd527	deploy 41min ago

receive
tx: 0x1db...db5a2

transact
19min ago

depositBalance
tx: 0xbc0...f69c8

transact
30min ago

addToArray
tx: 0x636...ca023

transact
32min ago

addToArray
tx: 0x4bb...70413

transact
32min ago

addToArray
tx: 0x95f...31ce5

transact
32min ago

number

Call

0:uint256: 10

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DEPLOY & RUN TRANSACTIONS

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0x5B3...eddC4

89.999 ETH

Deploy

Deployed Contracts

Transaction History

param0 (uint256)

Calldata

Params

Auto-Fill with AI Samples

owner

Call

0.address: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

signedNumber

Call

status

Call

student

Call

studentName

Call

Compile

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Exp 5.sol

1 // SPDX-License-Identifier: MIT

2 pragma solidity ^0.8.20;

3

4 contract Experiment5 {

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Scam Alert

Initialize as git repo

●

studentName

Call

0:string: Siddharth

REMIX

default_workspace

AI Code

DEPLOY & RUN TRANSACTIONS

Environment

Fork Reset

Remix VM

Cancun

Account 1

0x5B3...eddC4

89.999 ETH

Deploy

Deployed Contracts 1

Transaction History 15

signedNumber

Call

status

Call

student

Call

studentName

Call

0:string: Siddharth

Low level interaction

+

Value

0

wei

Gas limit

auto

0

29

30

31

32

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48

mapping(address => uint) public balances;

// =====

// 4. Struct

// =====

struct Student {

string name;

uint age;

bool registered;

}

Student public student;

// =====

// 5. Enum

// =====

Explain contract

AI copilot

0

Listen on all transactions

Filter with transaction hash or address

hash: 0x1f7...d6e14

Debug

CAL

[call] from: 0x5B38Da6a701c56854dCfc803Fc8875F56beddC4 to: Experiment5.studentName()

data: 0xb50...8ba31

Debug

REMIX

default_workspace

AI Code

DEPLOY & RUN TRANSACTIONS

Environment

Fork Reset

Remix VM

Cancun

Account 1

0x5B3...eddC4

89.999 ETH

Deploy

Deployed Contracts 1

Transaction History 15

numbers (uint256)

Call

param0 (uint256)

Calldata Params

Auto-Fill with AI Samples

owner

Call

0:address: 0x5B38Da6a701c56854dCfc803Fc8875F56beddC4

signedNumber

Call

status

Call

student

Call

1

2

3

4

5

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// SPDX-License-Identifier: MIT

pragma solidity ^0.8.20;

contract Experiment5 {

// =====

// 1. Primitive Data Types

// =====

uint public number = 10;

int public signedNumber = -5;

bool public isActive = true;

address public owner;

string public studentName = "Siddharth";

bytes32 public data = "Solidity";

// =====

Explain contract

AI copilot

0

Listen on all transactions

Filter with transaction hash or address

data: 0xb50...8ba31

Debug

CAL

[call] from: 0x5B38Da6a701c56854dCfc803Fc8875F56beddC4 to: Experiment5.owner()

data: 0x8da...5cb5b

Debug